



Autonomous Drone Navigation Using Computer Vision and Machine Learning

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Period 7 Syslab, 2026, Dr. Yilmaz

Demo



Introduction

Drones, or unmanned aerial vehicle that operates autonomously or controlled remotely, are used for a multitude of purposes including: filmmaking, delivery, search and rescue, and inspection work. They are especially useful because they are able to reach areas that are normally inaccessible or dangerous for people to access directly. However, many drones still rely on handheld controllers or pre-programmed instructions, which can make it less intuitive for users who want something quick and natural. For example, someone filming alone may find it difficult to capture smooth footage when trying to speak and fly the drone at the same time, or someone lacking in joint flexibility is unable to quickly do inputs required for drone actions. While drones are powerful tools, they do not adapt naturally to the movements or needs of the person using them. While the University of Tokyo may have created a drone that was able to precisely be controlled based on shoulder and hand positions using sensors applied to those areas, this approach still required additional technology on the user's body and did not solve the problem of minimizing equipment needed for drone movement. Furthermore, while preprogrammed instructions may work for predictable tasks, such as scanning a field for data, they are less effective in situations that require the drone to respond to sudden changes in the user's movement or environment.

Novelty

The novelty of our project is the use of real-time camera that register hand gestures as a means of more intuitive control as an alternative to traditional drone controllers. Instead of requiring a handheld controller or pre programmed flight path, the drone responds directly to the user's hand movements through a camera-based system. This makes the drone easier to operate for solo users, especially in filming or content creation, where controlling both the drone and the subject can be difficult. This project also explores how gesture-based control could improve accessibility into the drone space for users who may struggle with traditional controllers. By interpreting gestures in real time, the drone can respond more naturally to the user's intent.

Methods

i. Implementation

Our project intended to have three different modes of use: a hand gesture recognition mode, a hand tracking mode, and a machine learning portion. The drone autonomously switches modes after recognizing the Rockin' Pose through a camera using Computer Vision (Figure 1). The computer vision portions (gesture recognition and tracking mode) were implemented using both the drone's camera and an external camera for more applications. A user is able to control the computer vision using common devices, such as a computer camera or a phone camera. Computer vision connection to a phone is done using DroidCam.

ii. Computer Vision Portion:

We implemented OpenCV and MediaPipe to assist with our computer vision goals. MediaPipe allowed us to recognize the basic skeleton of our hands. Using MediaPipe, we compared pixel locations of different knuckles to analyze what gestures were being performed. We analyzed for a thumbs up (move up), thumbs down (move down), thumbs left (move left), thumbs right (move right), Rockin' pose (change mode), open palm (move forward), and closed palm (move backward) using a scaling method allowing for recognition at any distance from the camera.

iii. Machine Learning Portion

We designed a hoop using Fusion 360 using two concentric circles to serve as the edges of the hoop, one having a diameter of 27 inches and the other having a diameter of 24 inches. We laser cut the hoop out of wood strips.

After suspending the hoop mid air using two strings attached to thumbtacks on the wall, we took burst photographs and uploaded them to LabelStudio, where each was labeled with the appropriate direction the drone needed to travel to get through the hoop. There were nine different labels assigned: left, right, up, down, left down, left right, right down, right up, and forward.

To help our CNN identify only the ring's pattern, and not have it affected by background information, we decided to grayscale the entire image. For any pixel greater than 35 in black-intensity, we set the pixel to white. This helps make sure that the ring outline is the only thing visible by our CNN. This is seen in Figure 3, where grayscaling occurred on Figure 2.

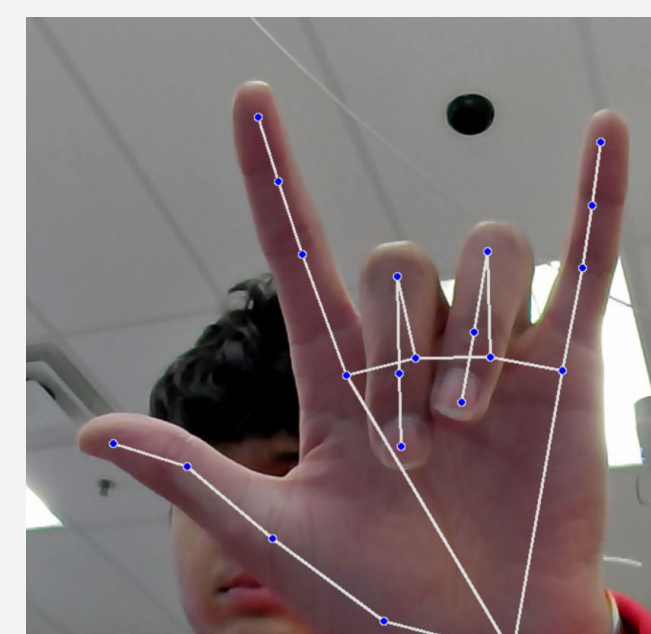


Figure 1 Rockin' Pose

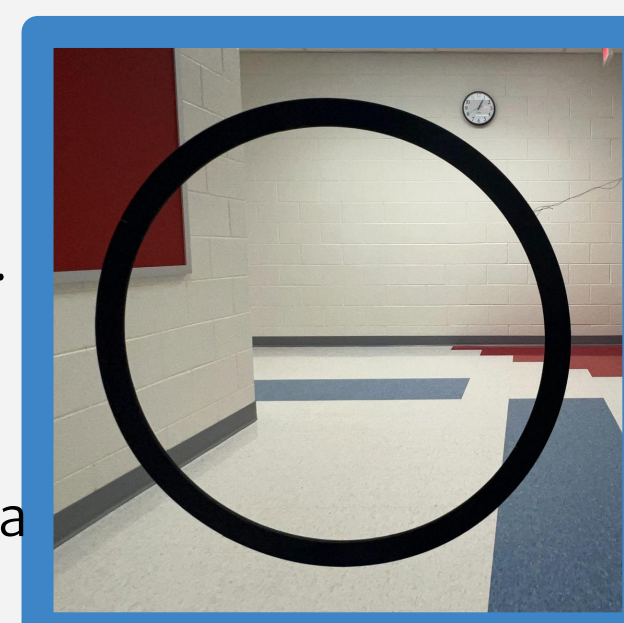


Figure 1 Permanent Hoop

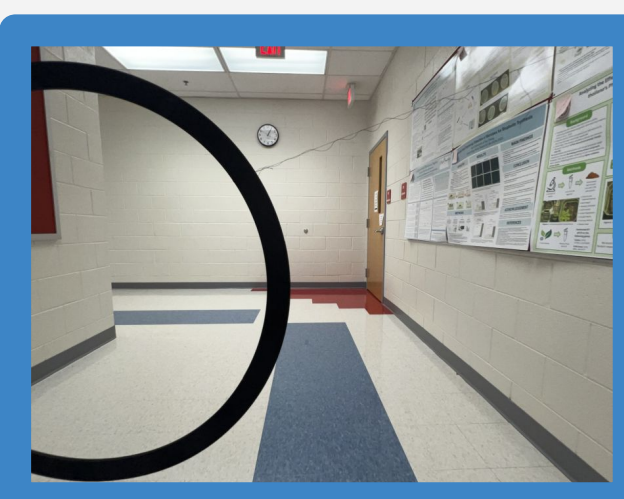


Figure 2 Normal Hoop Image

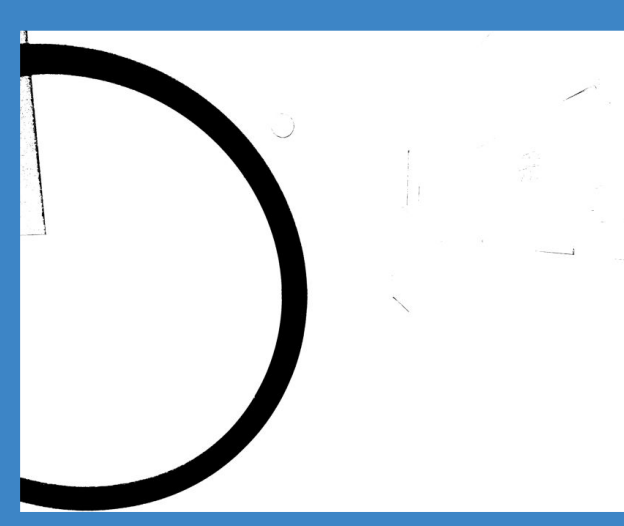
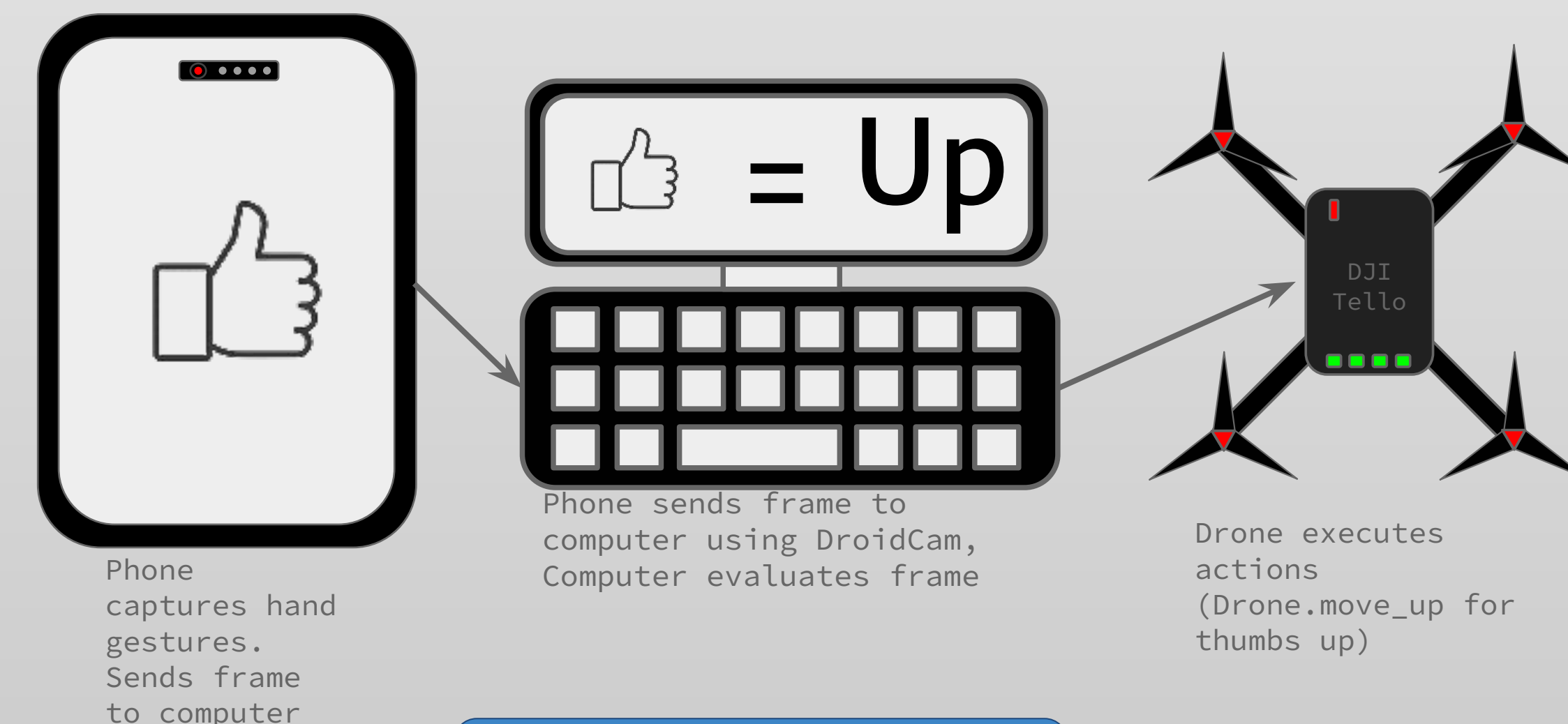


Figure 3 Greyscale Hoop Image



Results

Our convolutional neural network exhibited an accuracy of about 87% on a test dataset of 3035 images. Our computer vision system incorporated 8 different actions, ranging from moving in the cardinal directions, moving forward/backward, and switching navigation modes. In the end, our drone navigation system correctly executed all three different modes that we had set out to accomplish. The computer vision modes are very responsive and forgiving for newer pilots, and the machine learning portion successfully navigated through the ring on multiple test runs.

Future Work

We ventured into incorporating the Microsoft HoloLens into our project, where we would be able to control the drone using an external camera and see through its point of view. Unfortunately, due to time constraints, we were not able to implement the HoloLens fully into our setup, but our work provides a proper framework for doing so.

Through experimenting with the DJI Ryze Tello drone, we discovered a feature where the drone could estimate its height from the ground using infrared sensors. We discussed adding a LiDAR based system for ground elevation mapping but were not able to implement it.

Acknowledgements

We would like to thank Dr. Yilmaz for all of the guidance she has given us throughout this year-long project. Her passion for helping students was readily apparent throughout this whole process.



Autonomous Drone Control Using Gesture-Based Computer Vision and Convolutional Neural Network Navigation

Nick Boukens and Surin Wettimuny

Syslab – Yilmaz 2026

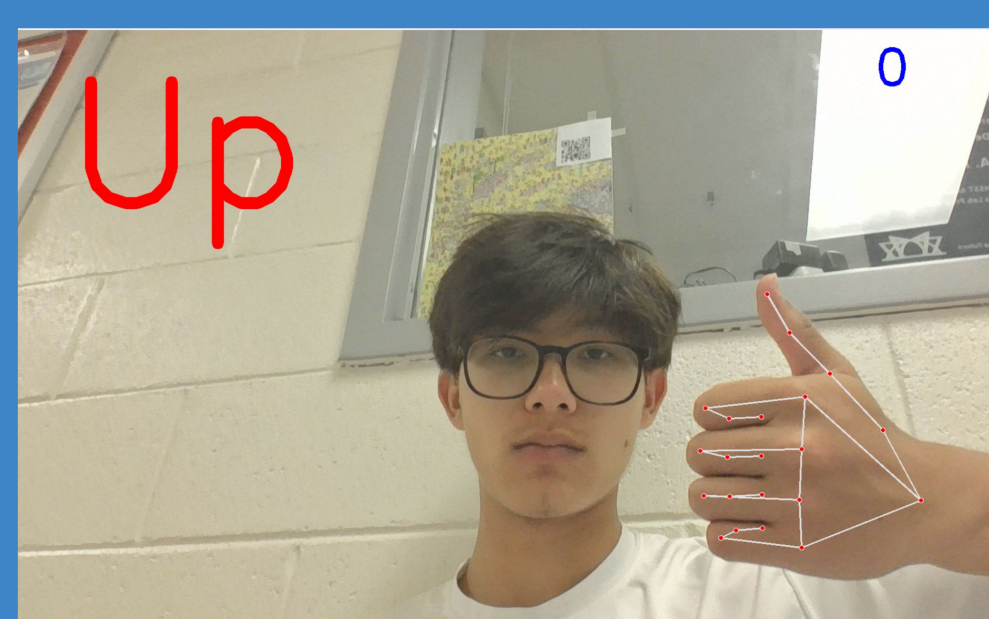


Introduction

Drones are defined as unmanned aerial vehicles that operate autonomously or are controlled remotely, and they are used for a multitude of purposes, including filmmaking, delivery, search and rescue operations, and inspection work. They are especially useful, as they are able to reach areas that are normally inaccessible or dangerous for people to access directly. However, many drones still rely on handheld controllers or pre-programmed instructions, which can make it less intuitive for users who want something quick and natural to use. For example, someone filming alone may find it difficult to capture smooth footage when trying to speak and fly the drone at the same time, or someone lacking in joint flexibility may be unable to quickly do inputs required for drone actions. Drones are powerful tools, but they do not adapt naturally to the movements or needs of the person using them. While the University of Tokyo may have created a drone that was able to precisely be controlled based on shoulder and hand positions using sensors applied to those areas, this approach still required additional technology on the user's body and did not solve the problem of minimizing equipment needed for drone movement. Furthermore, while preprogrammed instructions may work for predictable tasks, such as scanning a field for data, they are less effective in situations that require the drone to respond to sudden changes in the environment.

Novelty

The novelty of our project includes the use of real-time cameras that register hand gestures as a means of more intuitive drone control compared to other navigation options. Instead of requiring a handheld controller or pre-programmed flight path, the drone responds directly to the user's hand movements through a camera-based system. This makes the drone easier to operate for solo users, especially in filming or content creation, where controlling both the drone and the subject can be difficult. This project also explores how gesture-based control could improve accessibility into the drone space for users who may struggle with traditional controllers. By interpreting gestures in real time, the drone can respond more naturally to the user's intent.



Methods

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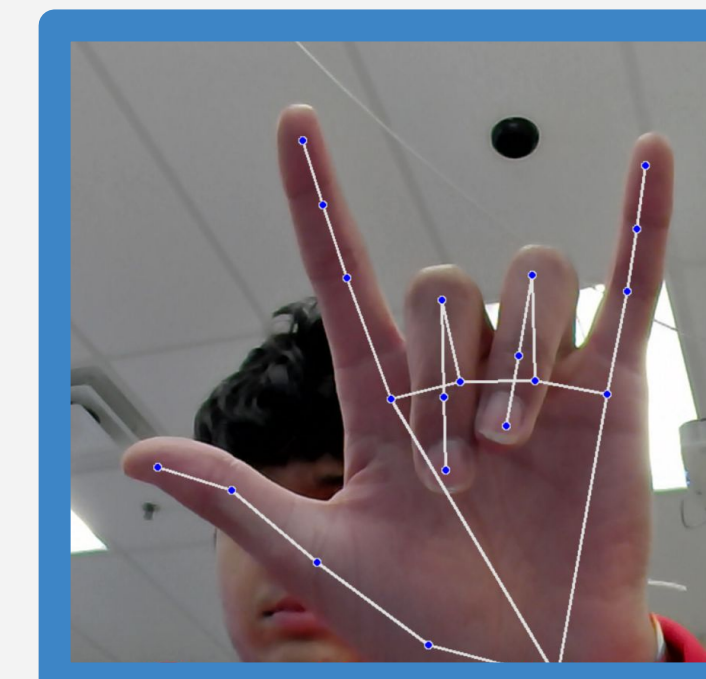


Figure 1 Rockin' Pose

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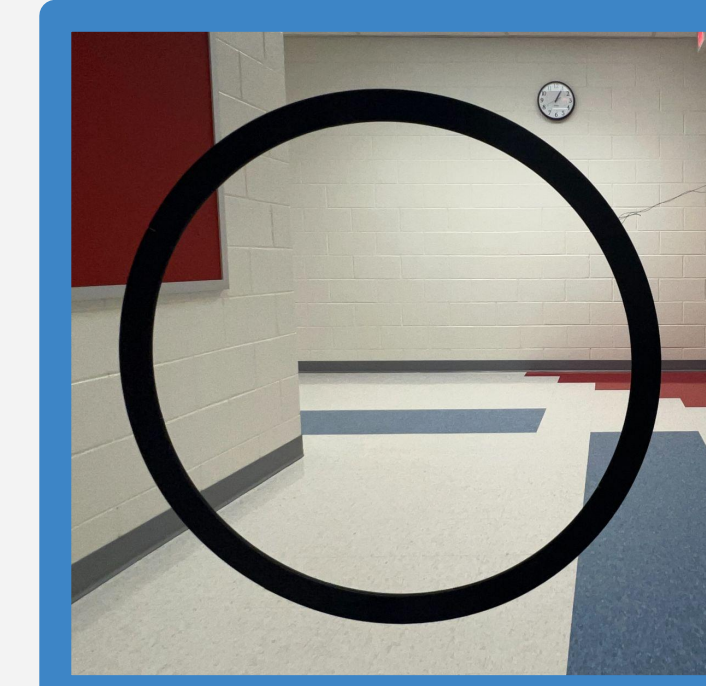


Figure 2 Permanent Hoop

iii. Machine Learning Portion

We designed a hoop, with Fusion 360, using two concentric circles to serve as the edges of the hoop, one having a diameter of 27 inches and the other having a diameter of 24 inches. We laser cut the hoop out of wood sheets. The final hoop is seen in Figure 2.

After suspending the hoop mid air using two strings attached to thumbtacks on the wall, we took burst photographs and uploaded them to LabelStudio, where each was labeled with the appropriate direction the drone needed to travel to get through the hoop. There were nine different labels assigned: left, right, up, down, left down, left right, right down, right up, and forward.

To help our CNN identify only the ring's pattern, and not have it affected by background information, we decided to grayscale the entire image. For any pixel greater than 35 in black-intensity, we set the pixel to white. This helps make sure that the ring outline is the only thing visible by our CNN. This is seen in Figure 4, where grayscaleing occurred on Figure 3.

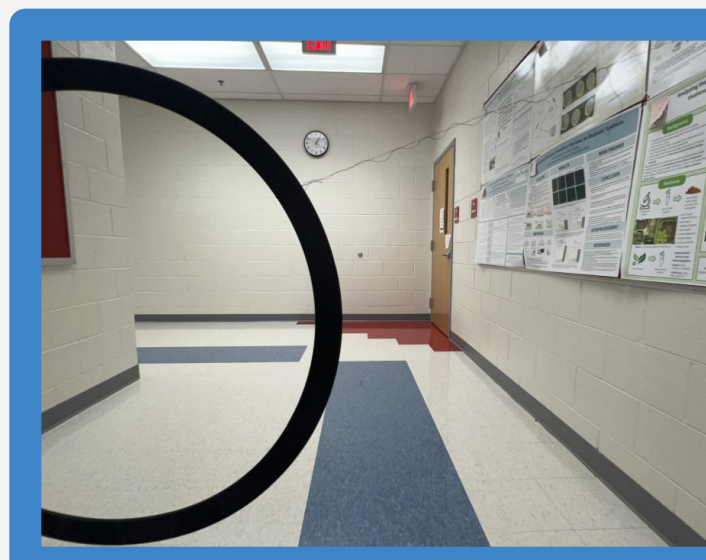


Figure 3 Normal Hoop Image

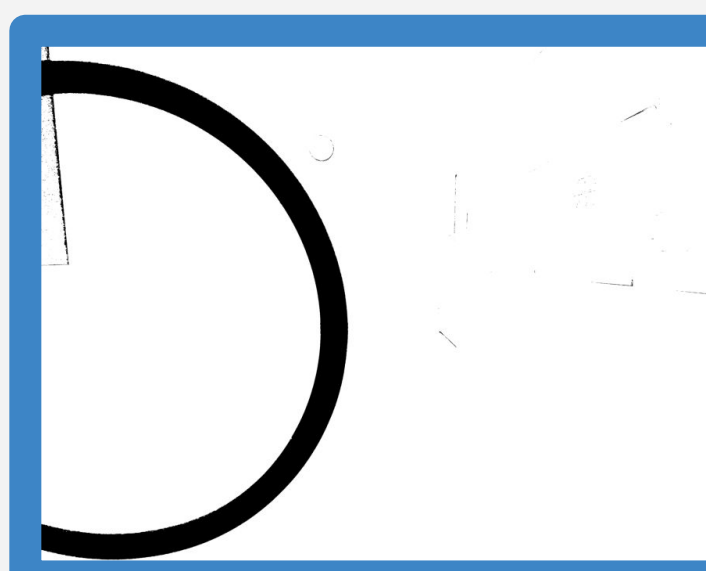
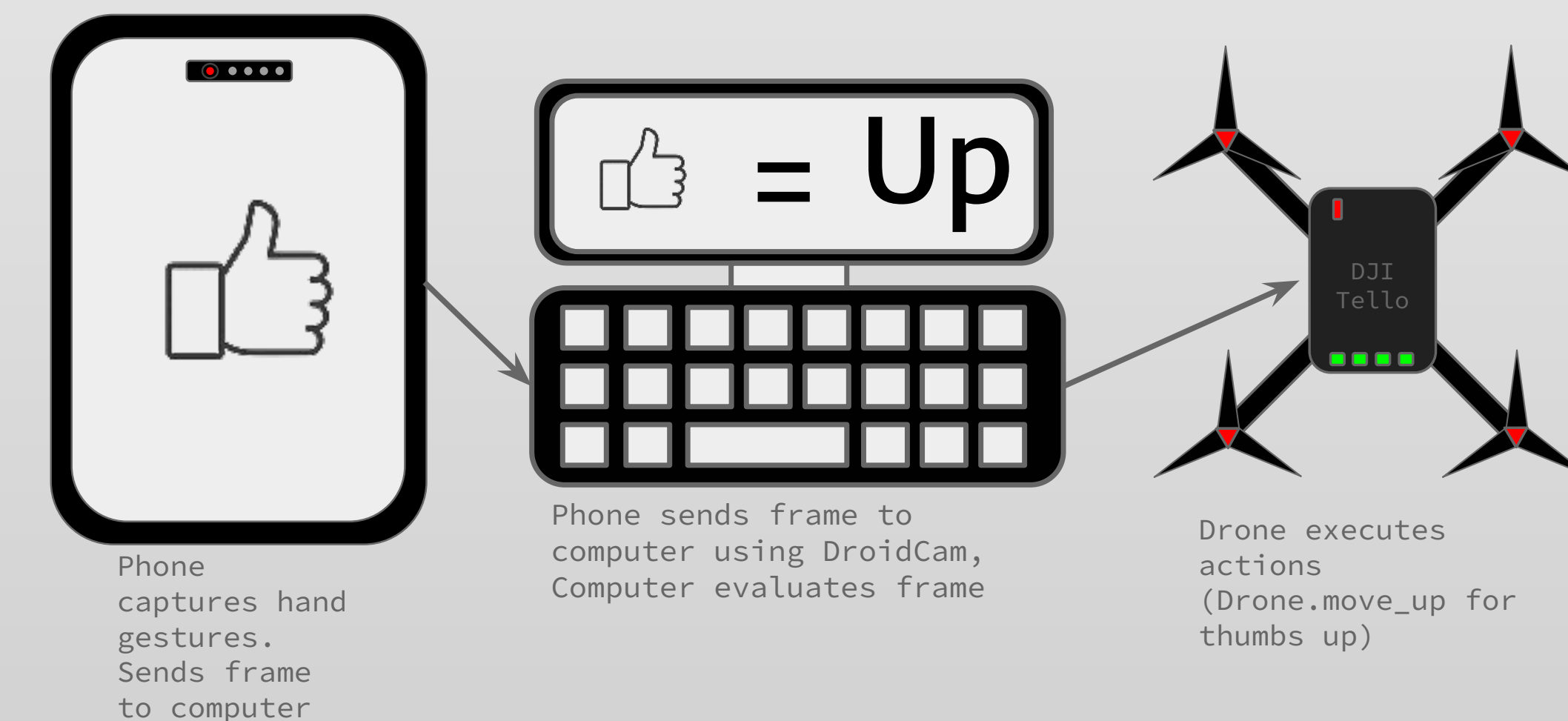


Figure 4 Grayscale Hoop Image



Results

Our convolutional neural network exhibited an accuracy of about 87% on a test dataset of 805 images. It was trained on 3224 images. Our computer vision system incorporated 8 different actions, ranging from moving in the cardinal directions, moving forward/backward, and switching navigation modes.

In the end, our drone navigation system correctly executed all three different modes that we had set out to accomplish. The computer vision modes are very responsive and forgiving for newer pilots, and the machine learning portion successfully navigated through the ring on multiple test runs.

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